Module-1

Syllabus

Accident Investigations and Risk Management: Collection of Accident Data, Assessment of Road Safety, Methods to Identify and Prioritize Hazardous Locations and Elements, Determine Possible Causes of Crashes, Crash Reduction Capabilities and Countermeasures, Effectiveness of Safety Design Features, Accident Reconstruction, Condition and Collision Diagram.

Introduction -Road safety Engineering-

Road Safety Engineering is the application of scientific and engineering principles to planning, design, operation, and maintenance of road infrastructure to reduce the risk, severity, and frequency of road crashes- Adapted from IRC: SP:88-2019, MoRTH Guidelines, and international best practices

Example: Using traffic survey data **(scientific)** and applying road geometry standards such as width & thickness **(engineering),** planners decide the road alignment location **(planning),** design proper curve radius **(design),** install traffic signals to reduce traffic jam and signs boards **(operation),** and repair potholes regularly **(maintenance)** to reduce accidents.

Definitions

Accident Investigation is the systematic process of collecting, analyzing, and documenting information related to a road crash to determine its causes, contributing factors, and preventive measures.

Risk Management in road safety refers to the identification, evaluation, and mitigation of potential hazards on roadways to reduce the chance and severity of future crashes.

Note:

- **Accident Investigation** =Find out what went wrong in a crash.
- **Risk Management** = Taking steps to stop it from happening again
- Studying why an accident happened, and then applying preventive steps to reduce the chances of it happening again.
- **Risk** = Chance that an accident might happen
- **Severity** = How serious the injuries or damage are when it happens
- **Frequency** = How often accidents happen at a location

Role of Engineering in Ensuring Road Safety

1. **Accident Prevention** – Prevents accidents by improving road design based on real data.
2. **User Protection** – Safeguards all road users including children, pedestrians, and drivers.
3. **Hazard Elimination** – Fixes dangerous spots using tested engineering solutions.
4. **Standard Compliance** – Ensures safer design by following IRC and MoRTH guidelines.
5. **Traffic Management** – Supports smooth traffic flow through signals, signage, and planning.
6. **Crash Cost Reduction** – Reduces cost and damage by preventing crashes before they happen.
7. **Sustainable Safety** – Maintains long-term safety through proper road maintenance.
8. **Night and Weather Safety** – Enhances safety in darkness and adverse conditions with lighting, reflectors, and high-friction surfaces.
9. **Public Confidence** – Builds trust in road networks, promoting safe mobility and economic growth.

Key Components of Road Safety Engineering

a) **Scientific Principles** = Speed affects braking distance → based on laws of motion and friction.
b) **Engineering Principles** = Technical guidelines engineers use to design and build roads that are safe, strong, and reliable for traffic.
c) **Planning** = Selecting Road location and features using topographic surveys, traffic forecasts, and land use analysis.
d) **Design** = Preparing detailed drawings with proper curves, lane widths, signals, signs, and pedestrian safety features as per standards.
e) **Operation** = Managing traffic flow using signals, signage, roundabouts, and speed regulations.
f) **Maintenance** = Ensuring safety by repairing potholes, repainting markings, and replacing damaged signs or lights.

Accident

The word "Accident" means an unforeseen and unplanned event that happens suddenly, often causing harm, injury, or damage, and occurring without deliberate intent.

In the road safety engineering context, an accident is an unexpected traffic event involving vehicles, pedestrians, or infrastructure that results in property damage, injury, or loss of life.

Collection of Accident Data

Collecting accurate accident data is the first and most crucial step in road safety analysis. It helps investigators understand the frequency, causes, and severity of crashes, identify hazardous locations, and plan effective countermeasures. The following points show the **step-by-step process of accident data collection**:

1. **Visit the Location**
   - Physically inspect the exact accident-prone site (e.g., sharp curve, junction, work zone).
   - Note geometric features, visibility conditions, signage, lighting, traffic volume, and surrounding land use.
2. **Check Police Records**
   - Collect First Information Reports (FIRs) and accident case files from the nearest police station.
   - These records provide official details such as time of accident, type of vehicles involved, number of casualties, and possible causes.
3. **Use Hospital Data**
   - Gather information from hospitals or trauma care centres regarding injured or deceased victims from that location.
   - This provides insights into accident severity and helps cross-check police records.
4. **Install or Review CCTV Footage**
   - Check existing surveillance cameras or install temporary ones.
   - Video evidence helps in understanding traffic behaviour, pedestrian movement, speeding, and violation patterns at the time of crashes.
5. **Interview Locals or Witnesses**
   - Speak with shopkeepers, residents, frequent drivers, or pedestrians who use the spot regularly.
   - Eyewitness accounts give context to recurring issues (e.g., blind curves, over speeding, poor lighting).
6. **Fill Accident Data Forms**
   - Use standard formats such as IRC accident data forms.
   - Record essential details: date, time, weather, road surface condition, type of collision, type of vehicles, driver behaviour, and casualties.
7. **GPS Mapping**
   - Use GPS or GIS to mark exact coordinates of accident locations.
   - Mapping helps in creating accident black spot maps, cluster analysis, and trend visualization.
8. **Photograph the Scene**
   - Take clear photographs of skid marks, damaged guardrails, faulty signs, potholes, or missing markings.
   - Visual evidence supports technical analysis and is useful in presentations or reports.

Post-Accident Data Collection

1. **Analyse Crash Data**
    - Review all collected accident reports, police records, hospital data, and field observations.
    - Identify recurring patterns such as peak time of crashes (day/night), vehicle category (car, bus, truck, two-wheeler), weather conditions (rain, fog), and collision type (rear-end, run-off-road, head-on).
2. **Identify Black Spots**
    - Pinpoint locations with unusually high accident frequency or severity.
    - Use crash mapping (GIS/GPS) to visually highlight clusters of crashes and accident-prone curves, intersections, or junctions.
3. **Prioritize Locations**
    - Rank accident-prone spots based on measurable indicators such as:
        - Crash Rate (accidents per million vehicle-km).
        - Severity Index (fatal + serious crashes per total crashes).
        - Hazard Scoring Method (weighted score for fatalities and serious injuries).
    - This helps focus on sites with the greatest safety impact.
4. **Conduct Road Safety Audit (RSA)**
    - Perform a structured safety review using the IRC: SP:88 Road Safety Audit checklist.
    - Check for deficiencies in road geometry, sight distance, signage, lighting, pavement condition, and pedestrian safety.
    - Conduct both day and night audits for complete assessment.
5. **Propose Safety Measures**
    - Suggest corrective actions tailored to the identified issues. Examples:
        - Low-cost measures: Installing chevron boards, speed limit signs, reflective studs.
        - Medium measures: Improving curve radius, road widening, additional lighting.
        - High-cost measures: Grade separation, bypasses, or full intersection redesign.
6. **Prepare Detailed Report**
    - Document all findings, accident diagrams, photographs, condition and collision diagrams.
    - Provide engineering drawings, cost estimates, and justifications for proposed safety measures.
    - Ensure the report follows standard formats for submission to authorities.
7. **Estimate Cost & Get Approval**
    - Work out the financial requirements for the proposed improvements.
    - Submit the proposal to funding agencies or road authorities for technical and financial clearance.

8. **Implement Engineering Solutions**
    - Execute corrective actions at the site such as installing traffic control devices, resurfacing roads, or redesigning hazardous curves.
    - Coordinate with contractors, traffic police, and local authorities during implementation.
9. **Monitor Impact**
    - After implementation, monitor accident data for 6–12 months to check effectiveness.
    - Compare pre- and post-implementation crash statistics to measure accident reduction.
    - Fine-tune measures if results are not satisfactory.

Assessment Of Road Safety

Assessment means examining a road or junction to find out how safe it is, and identifying what improvements are needed.

Assessment of road safety is a systematic evaluation of the physical condition, design features, and operational characteristics of a road or traffic facility to identify existing or potential safety hazards for all road users. *(Adapted from IRC: SP:88-2019 and international road safety audit practices)*

Steps to Assess Road Safety

1. **Select the Spot** – Choose accident-prone locations (black spots), high-traffic junctions, school zones, or areas flagged by road users. This ensures focus on critical points with safety risks.
2. **Collect Background Information** – Gather available records such as accident statistics, traffic volume counts, design drawings, and previous road safety audits. This helps understand the history and importance of the location.
3. **Site Inspection** – Visit the site physically to observe existing road geometry, pedestrian crossings, signage, markings, street lighting, and driver/ pedestrian behavior during both day and night.
4. **Identify Deficiencies –** Note all safety issues such as faded markings, potholes, missing signs, inadequate lighting, poor visibility at curves, drainage problems, and encroachments.
5. **Use Audit Checklist** – Apply a structured safety audit checklist (as per IRC: SP:88 Road Safety Audit guidelines) to ensure that no critical aspect is overlooked.
6. **Classify Risk** – Categorize issues into low, medium, or high risk based on their potential to cause accidents. For example, a missing speed breaker in a school zone would be high risk.
7. **Suggest Solutions** – Recommend engineering and management measures like speed breakers, guardrails, traffic signals, zebra crossings, warning signboards, pedestrian refuges, or curve realignment.

8. **Document & Submit Report** – Prepare a structured report with photographs, sketches, detailed observations, proposed solutions, and cost estimates. Submit it to the concerned road authority.
9. **Monitor Post-Action** – After corrective measures are implemented, revisit the site periodically to analyse whether crashes have reduced and safety has improved.

Note:

1. **Accident Data Collection at Spot:** Recording exact details and evidence immediately at the accident location.
2. **Post-Accident Data Collection:** Gathering and analyzing accident reports over time to improve safety.
3. **Steps to Assess Road Safety:** Checking roads for problems to prevent accidents before they happen

Methods to Identify & Prioritize Hazardous Locations

Hazardous locations on roads are identified to reduce crash risks and improve safety. These methods use accident data, field inspections, and policy guidelines to find and rank black spots. The aim is to focus resources on the most critical locations where safety measures are urgently needed. These methods are classified based on the approach used—whether it relies on statistical data, field observations, or official policy criteria, as shown below:

a) **Data-based Methods:**
1. Accident Frequency Method
2. Accident Rate Method
3. Accident Severity Method
4. Weighted Hazard Index (WHI)

b) **Visual / Inspection-based Methods:**
1. GIS-Based Spatial Mapping
2. Road Safety Audit (RSA)

c) **Policy / Guideline-based Methods:**
1. Black Spot Criteria (MoRTH Guidelines)

The following table gives highlights on various methods used to identify and prioritize hazardous road locations and their key outcomes.

SL NO	Method	What it Finds Out
1	Accident Frequency Method	Finds roads with the highest number of crashes.
2	Accident Rate Method	Finds how many crashes happened but also where traffic is more dangerous.
3	Accident Severity Method	Total number of accidents and gives weightage based on seriousness (fatal/minor).
4	Weighted Hazard Index (WHI)	Finds Hazard Index by adding- (count score + severity score + exposure score).

5	GIS-Based Spatial Mapping	Finds accident cluster spots on a map.
6	Road Safety Audit (RSA)	Finds visible safety issues by site inspection.
7	Black Spot Criteria (MoRTH)	Finds officially recognized black spots (10+ severe crashes in 3 yrs).

Table 1.1: Road Safety Hazard Analysis Methods

Explanation

1.Accident Frequency Method

The Accident Frequency Method identifies road locations with the highest number of crashes over a specific period. It is a data-based approach that focuses purely on the count of accidents, without considering traffic volume or severity. By highlighting spots with repeated accidents, engineers can prioritize these areas for detailed investigation and corrective safety measures. This method is simple to apply and is often used as an initial screening tool to detect black spots on roads.

Highlights of Accident frequency method

- Identifies locations with the highest number of crashes over a period.
- Data-based method focusing on crash counts only.
- Simple to apply and good for initial screening of black spots.
- Does not account for traffic volume or accident severity.
- Helps engineers prioritize sites for detailed safety analysis.

Example:

- Road A: 20 accidents/year
- Road B: 15 accidents/year
- **Inference: Road A is critical**, because it has the **higher number of accidents**.

2.**Accident Rate Method**

The Accident Rate Method identifies road locations where crashes occur relative to traffic volume, rather than just total crash count. It is a data-based approach that calculates the number of accidents per million vehicle-kilometres travelled, providing a better understanding of where the road is most dangerous for users. This method helps prioritize sites that may have fewer crashes but higher risk due to heavy traffic.

Accident Rate Method – Highlights

- Measures crash relative to traffic volume (crashes per million vehicle-km).
- Data-based method giving a normalized risk assessment.

- Identifies roads that are dangerous even if total crashes are low.
- Helps prioritize sites more accurately than frequency alone.
- Supports engineers in targeting high-risk locations for safety improvements.

Formula to calculate Accident rate method

$$\text{Accident Rate} = \frac{\text{Number of Accidents in a Year}}{\text{Traffic Volume per Day} \times 365} \times 10^6$$

Example:

Road	Accidents in 1 year	Traffic volume (vehicles/day)	Vehicles in a year	Accident Rate (per million vehicles)
A	40	20,000	7,300,000	5.48
B	30	10,000	3,650,000	8.22

Inference: Road B has a higher accident rate per vehicle, means higher risk per vehicle, even though the total number of accidents is less.

3.Accident Severity Method

The Accident Severity Method evaluates road locations based on the seriousness of crashes, not just the total number. It assigns weightage to accidents depending on their severity—fatal, serious injury, or minor injury. This data-based approach helps identify sites where even a few accidents can have a high impact, ensuring that areas with severe crashes are prioritized for safety improvements**.**

Accident Severity Method – Highlights

- Focuses on the seriousness of accidents (fatal, serious, minor).
- Data-based method giving weighted importance to crashes.
- Helps prioritize locations needing urgent safety measures.
- Uses a Severity Index to quantify the impact of accidents.
- Severity Index is a numerical measure of how serious accidents are at a location.
- Higher SI (Severity Index) → indicates that accidents at that location are generally more serious, with a greater proportion of fatal or major injury crashes.
- Lower SI (Severity Index) → indicates that accidents are mostly minor, with fewer serious or fatal injuries.

$$SI = \frac{(Fatal \times 3) + (Serious \times 2) + (Minor \times 1)}{\text{Total Accidents}}$$

Example

Road	Fatal	Serious	Minor	Total Accidents
A	2	4	10	16
B	1	5	12	18

Severity Index (SI) using 3-2-1 weights:

$$SI_A = \frac{(2 \times 3) + (4 \times 2) + (10 \times 1)}{16} = \frac{6 + 8 + 10}{16} = \frac{24}{16} = 1.5$$

$$SI_B = \frac{(1 \times 3) + (5 \times 2) + (12 \times 1)}{18} = \frac{3 + 10 + 12}{18} = \frac{25}{18} \approx 1.39$$

Road A has a higher severity index (1.5) compared to Road B (1.39). Even though Road B has more total accidents, the average severity is lower.

4.Weighted Hazard Index (WHI)

The Weighted Hazard Index (WHI) is a composite measure used to evaluate hazardous locations by combining multiple factors—accident frequency, severity, and exposure (traffic volume). Each factor is assigned a score, and their sum gives a hazard index for the site. This method allows engineers to prioritize black spots more effectively than using any single factor alone, as it considers both the number and seriousness of accidents along with traffic exposure.

Weighted Hazard Index (WHI) – Highlights

- Combines accident frequency, severity, and traffic exposure into a single score.
- Data-based method providing a comprehensive risk assessment.
- Assigns higher scores to locations with severe and frequent crashes on busy roads.
- Helps in ranking hazardous sites objectively.
- Supports engineers in allocating resources to the most critical black spots.

- WHI gives more weight to fatal, less to serious, and least to minor accidents.
- Measures: Combines frequency, severity, and traffic exposure into one score.
- Provides a comprehensive risk measure to prioritize hazardous locations.
- Balances on how many accidents happen? how serious they are? and how much traffic is exposed? giving the real picture of risk.

Example

- **Road A:** Fatal = 1, Serious = 3, Minor = 6, **ADT** = 12,000 vehicles/day
- **Road B:** Fatal = 0, Serious = 4, Minor = 8, **ADT** = 6,000 vehicles/day

- Consider (weights: Fatal=3, Serious=2, Minor=1)

List of Formulas used

1. **Frequency (count)** = Fatal + Serious + Minor
2. Severity Score (3–2–1) = $3 \times \text{Fatal} + 2 \times \text{Serious} + 1 \times \text{Minor}$
3. **Annual Exposure** = ADT × 365
4. **Hazard Index** = Frequency (Count) + Severity Score + Annual Exposure

Road	Fatal	Injury	Minor	Frequ -ency count	Severity Score	ADT	Annual Exposure (million veh/yr)	Hazard Index = Count + Severity + Exposure
A	1	3	6	10	3*1 + 2*3 + 1*6) = **15**	12,000	12,000×365 4,380,000 = **4.38 million**	**10 + 15 + 4.38= 29.38**
B	0	4	8	12	3*0 + 2*4 + 1*8) = **16**	6,000	6,000×365 2,190,000 =**2.19 million**	**12 + 16 + 2.19= 30.16**

5.GIS-Based Spatial Mapping

GIS-Based Spatial Mapping uses geographic information systems (GIS) to visually analyse and identify accident-prone areas on roads. By plotting accident locations on a map, clusters or patterns of crashes become apparent, helping engineers detect high-risk spots efficiently. This visual/inspection-based method allows for spatial analysis of factors like road geometry, traffic volume, and surrounding land use, supporting data-driven decision-making for road safety improvements. GIS is tool that uses digital maps to show where crashes happen, making it easier to spot accident-prone areas

GIS-Based Spatial Mapping – Highlights

- Uses GIS tools to map and visualize accident locations.
- Visual/inspection-based method highlighting clusters and patterns.
- Helps identify high-risk spots efficiently.
- Supports spatial analysis of road features, traffic, and environment.
- Assists engineers in prioritizing black spots for safety interventions.

6.Road Safety Audit (RSA)

A Road Safety Audit (RSA) is a systematic on-site inspection of roads to identify safety issues and potential hazards before or after accidents occur. Conducted by a team of qualified experts, it evaluates road geometry, signage, markings, lighting, pedestrian facilities, and traffic behaviour. This visual/inspection-based method helps detect deficiencies that may not appear

in accident records, allowing engineers to recommend practical safety improvements and prevent crashes proactively.

Road Safety Audit (RSA) – Highlights

- Systematic field inspection of roads by safety experts.
- Visual/inspection-based method identifying hazards not visible in data.
- Checks road geometry, signage, markings, lighting, and pedestrian facilities.
- Helps detect deficiencies before or after accidents occur.
- Supports engineers in proposing practical safety improvements**.**

.7. Black Spot Criteria (MoRTH Guidelines0

The Black Spot Criteria, as defined by MoRTH guidelines, identifies road locations that are officially recognized as hazardous due to repeated accidents. A site qualifies as a black spot if it records 10 or more fatal/serious accidents within three years. This policy/guideline-based method helps authorities prioritize road safety interventions and allocate resources to sites with the greatest need for improvement**.**

Black Spot Criteria (MoRTH Guidelines)

- Identifies locations with 10+ fatal/serious accidents in 3 years.
- Policy/guideline-based method for official recognition of black spots.
- Focuses on sites with repeated high-severity crashes.
- Helps authorities prioritize safety interventions.
- Supports resource allocation to the most critical locations.

Possible Causes of Crashes

Crashes occur due to a combination of road, traffic, driver, vehicle, environmental, and pedestrian factors. Understanding these causes helps engineers and authorities plan effective countermeasures to reduce accidents.

a) **Road Design Issues** – Poor geometric or structural design can make it difficult for drivers to navigate safely.
   Example: Sharp curves, sudden lane merges, inadequate shoulders, or poor sight distance.
b) **Traffic Control Issues** – Ineffective traffic management and unclear control devices lead to confusion and unsafe driving.
   Example: Missing or faded road signs, non-working traffic signals, unclear lane markings.
c) **Driver Behaviour** – Human errors or risky driving practices are among the leading causes of crashes.
   Example: Over-speeding, drunk driving, wrong-side driving, distracted driving, fatigue.

d) **Vehicle Condition** – Mechanical or maintenance failures can cause sudden loss of control.
   Example: Brake failure, tyre blowout, steering problems, worn-out suspension.
e) **Environmental Conditions** – Weather and lighting conditions affect visibility and road surface friction.
   Example: Rain, fog, snow, darkness, low visibility due to dust or smoke.
f) **Pedestrian Factors** – Unsafe pedestrian actions or lack of proper pedestrian facilities contribute to accidents.
   Example: Jaywalking, no zebra crossings, absence of footpaths, or missing pedestrian signals.

Crash Reduction Capability (CRC)

Crash Reduction Capabilities (CRC) indicate the effectiveness of specific interventions in reducing the number or severity of road accidents. It measures how much a safety measure—such as installing guardrails, improving street lighting, or adding speed cameras—can lower crash likelihood or impact, often expressed as a percentage reduction in crashes or injuries. CRC helps engineers and planners prioritize measures that provide the greatest safety benefit.

Countermeasures

Countermeasures are the actual actions implemented to enhance road safety. These include engineering measures (e.g., road widening, rumble strips), enforcement strategies (e.g., speed limits, drunk-driving checks), and educational initiatives (e.g., awareness campaigns, driver training). They are selected based on crash data, road conditions, and user behaviour to prevent accidents and reduce their severity effectively**.**

Crashes on roads can be reduced through a combination of engineering, enforcement, and educational interventions. Understanding and implementing these countermeasures helps improve safety, prevent accidents, and protect all road users.

Countermeasures to Improve Road Safety

1. **Engineering Countermeasures**
   • Road Design Improvements – Widen lanes, provide proper banking on curves, improve sight distance.
   • Traffic Control Devices – Install signs, signals, speed breakers, rumble strips, and reflective markers.
   • Roadside Safety Features – Guardrails, crash barriers, pedestrian crossings, and medians to reduce severity.
   • Lighting and Visibility Enhancements – Streetlights, reflective paint, and high-visibility markings at critical locations.
2. **Enforcement Countermeasures**
   • Speed Control – Strict enforcement of speed limits with cameras or patrols.
   • Drunk Driving Checks – Random alcohol checks and penalties for violations.
   • Helmet and Seatbelt Laws – Mandatory compliance with fines for non-use.

• Traffic Rules Enforcement – Strict action against wrong-side driving, illegal overtaking, and red-light violations.

3. **Educational Countermeasures**
   • Public Awareness Campaigns – Promote safe driving habits via media, workshops, and road safety weeks.
   • Driver Training Programs – Emphasize defensive driving and hazard perception skills.
   • School and Community Education – Teach children and pedestrians about road safety rules.
   • Signage Interpretation Education – Educate Road users to understand traffic signs, signals, and road markings correctly.

Effectiveness of Safety Design Features-Road safety Features

Road safety features are designed to prevent accidents, reduce crash severity, and protect road users. Their effectiveness can be evaluated based on how well they control vehicle movement, provide warnings, and minimize injury in case of a crash. Common features and their effectiveness include:

1. **Guardrails and Crash Barriers** – Prevent vehicles from leaving the roadway or hitting fixed objects; reduce severity of run-off-road crashes.
2. **Rumble Strips** – Alert drivers to upcoming curves or lane departures, reducing speed-related accidents.
3. **Signage and Pavement Markings** – Provide clear guidance and warnings; reduce confusion and collisions at intersections or hazardous sections.
4. **Median Dividers** – Separate opposing traffic, reducing head-on collisions.
5. **Pedestrian Crossings and Footpaths** – Protect pedestrians, reduce pedestrian-vehicle crashes.
6. **Lighting and Reflectors** – Improve visibility at night or during poor weather, reducing night-time crashes.

Accident reconstruction

Accident reconstruction in road safety engineering is the systematic process of investigating and scientifically recreating a road crash to determine its causes, contributing factors, and sequence of events.

It is used to understand why the accident occurred and to design countermeasures that improve overall road safety.

Key Aspects of Accident Reconstruction

- **Site Inspection & Data Collection** – Gather skid marks, vehicle positions, damage, road geometry, lighting, and weather conditions.
- **Vehicle Dynamics Analysis** – Estimate speed, braking distance, impact angle, and loss of control.

- **Human Factors** – Study driver perception, reaction time, fatigue, or impairment.
- **Environmental & Road Factors** – Check for poor signage, low friction surface, inadequate lighting, or faulty traffic control.
- **Simulation & Modelling** – Use software/tools to recreate crash scenarios.
- **Reporting** – Provide evidence-based explanation and recommend safety measures.

Condition And Collision Diagram

In road safety engineering, understanding how and why an accident occurred is crucial to prevent future crashes. Condition and Collision Diagrams are drawn to visually represent the accident scene, vehicle movements, and collision dynamics, helping engineers and investigators analyse contributing factors, identify hazardous locations, and recommend corrective measures.

Definitions

- **Condition Diagram** – A condition diagram is a scaled sketch of the crash location showing road geometry, traffic control devices, landmarks, and physical features. It provides a clear picture of the environment where accidents occur.
- **Collision Diagram** – A collision diagram is a graphical representation that shows how and where crashes happened at an intersection or road section, using standardized symbols to indicate the type, direction, and severity of collisions.

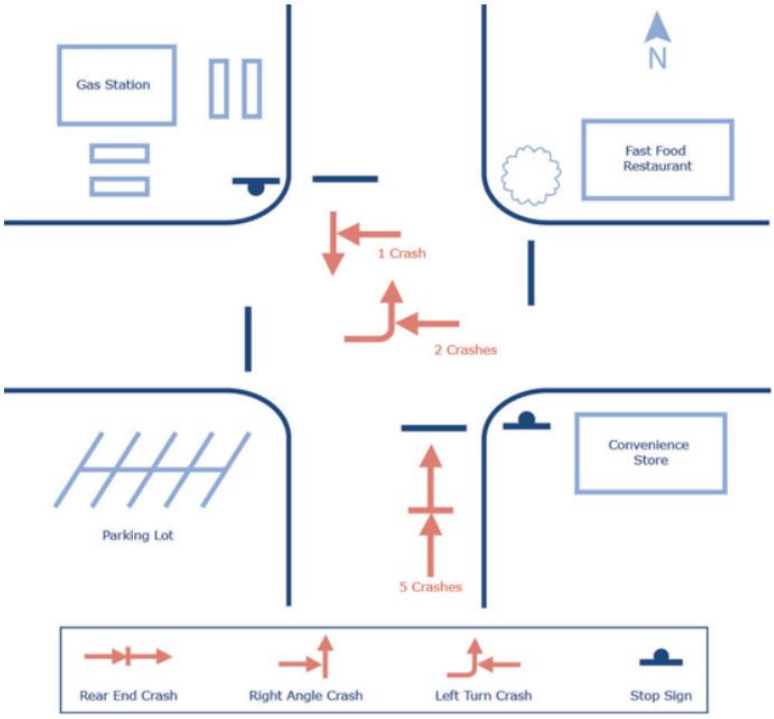

Figure 1.1: Collision diagram drawn on Condition diagram

Condition Diagram-Explanation

Condition Diagram – Explanation

A Condition Diagram is a visual representation of the roadway, surrounding environment, and conditions at the time of a crash. It shows elements such as lane markings, road width, intersections, traffic signs, signals, road furniture, lighting, weather, and surface conditions. This diagram helps investigators and engineers understand the context in which the accident occurred, identify hazardous factors, and plan effective countermeasures. It is particularly useful for analysing patterns in accidents and designing safer roads.

Condition Diagram – Highlights

- Visual representation of road, environment, and crash conditions.
- Shows lane markings, intersections, road width, signs, signals, and traffic control devices.
- Includes weather, lighting, and surface conditions at the time of the accident.
- Helps investigate contributing factors of crashes.
- Supports planning of corrective and preventive measures.

Collision diagram-Explanation

A Collision Diagram is a graphical representation of the movement and interaction of vehicles involved in a crash. It illustrates the direction, speed, point of impact, and final positions of all vehicles at the accident scene. This diagram helps investigators reconstruct the sequence of events, determine the causes of collisions, and evaluate contributing factors such as driver behaviour, vehicle condition, and road layout. Collision diagrams are essential for analysing crash dynamics and recommending targeted safety measures.

Collision Diagram – Highlights

- Graphically shows vehicle movements and interactions during a crash.
- Indicates direction, speed, point of impact, and final positions of vehicles.
- Helps in reconstructing the sequence of events.
- Assists in identifying causes and contributing factors of collisions.
- Supports designing targeted countermeasures to prevent future accidents.

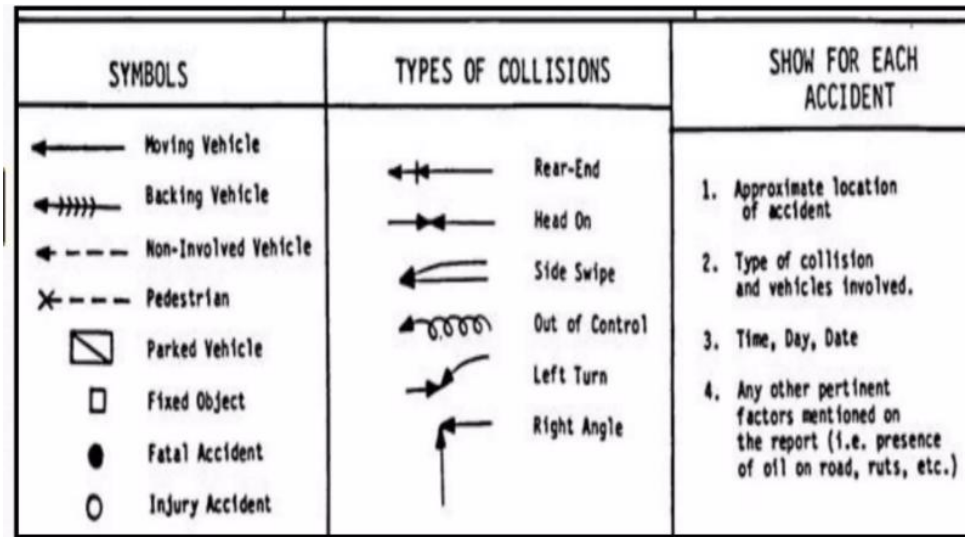

Figure 1.2: Symbols and type of Collisions

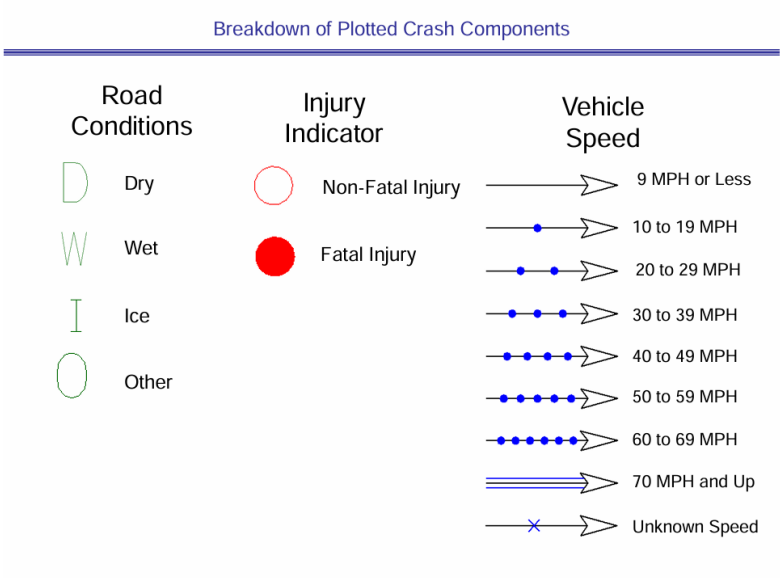

Figure 1.3: Crash component symbols

Relationship Between Condition and collision diagram

- **Condition Diagram** – A **static representation** showing road geometry, signs, markings, lighting, roadside objects, gradients, and visibility.
- **Collision Diagram** – A **dynamic representation** illustrating vehicle paths, points of collision, angle of impact, and vehicle movements just before and after the crash.
- Condition Diagram = road map
- Collision diagram = condition diagram + crash points+ **symbols + type of collision**.

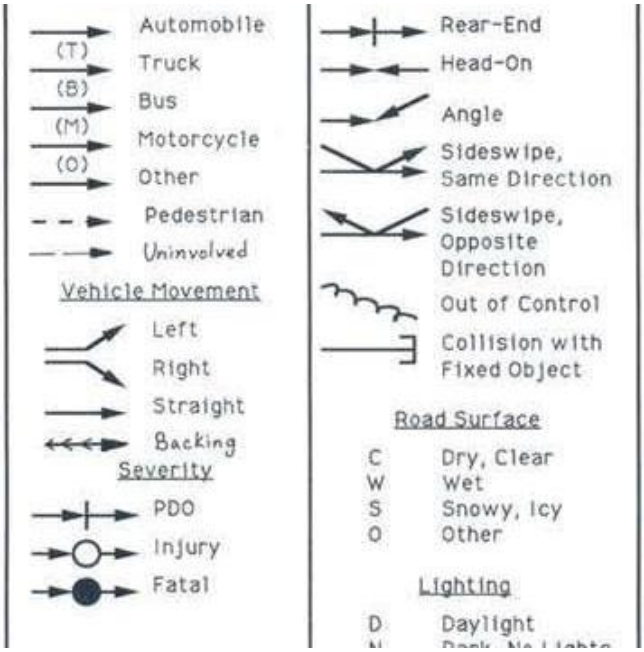

Figure 1.4: Crash component symbols